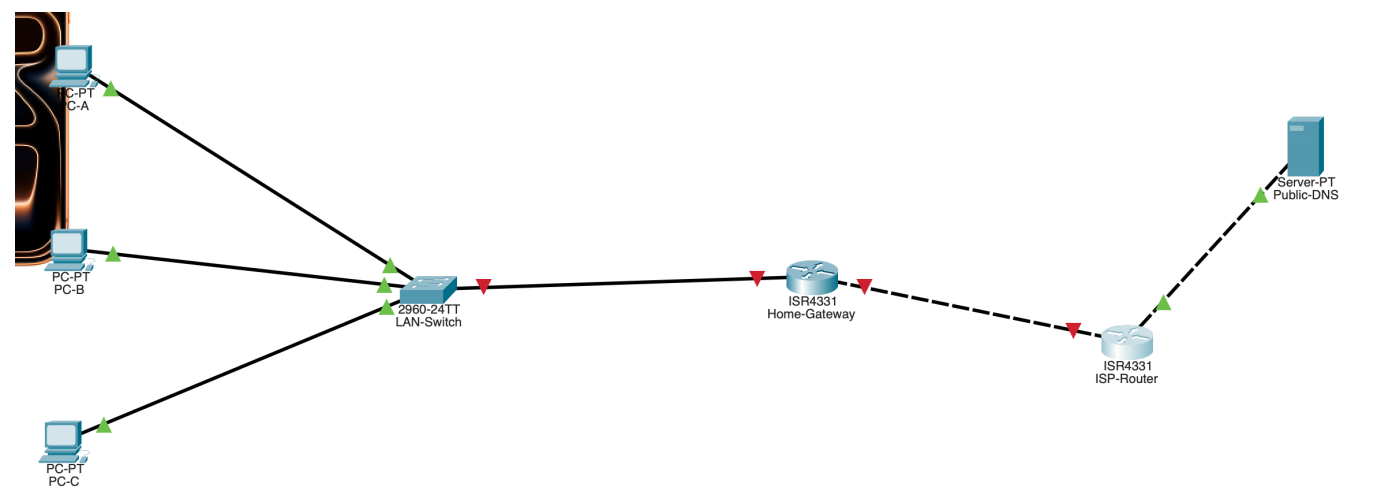


**Lab 09****Course:** Networks System Design**Name:** Do Davin**Student ID:** P20230018**Instructor:** Mr. Kuy Movsun**Due Date:** Tuesday, 30 December 2025, 12:00 AM

1. Topology & Addressing



Public-DNS

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

X

IP Configuration

DHCP

Static

IPv4 Address

8.8.8.8

Subnet Mask

255.255.255.0

Default Gateway

8.8.8.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::230:F2FF:FE76:DD48

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

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Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

169.254.166.23

Subnet Mask

255.255.0.0

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::230:A3FF:FEEB:A617

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

2. Pre-Lab Setup (Instructor Configuration)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP-Router
ISP-Router(config)#interface g0/0/1
ISP-Router(config-if)#ip address 8.8.8.1 255.255.255.0
ISP-Router(config-if)#no shutdown

ISP-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

ISP-Router(config-if)#exit
ISP-Router(config)#interface g0/0/0
ISP-Router(config-if)#ip address 203.0.113.1 255.255.255.252
ISP-Router(config-if)#no shutdown

ISP-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

ISP-Router(config-if)#end
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#
```

3. Implementing NAT (Network Address Translation)

Step 3A: Configure WAN Link & Default Route

```
Router>enable
Router#
Router#|
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname Home-Gateway
Home-Gateway(config)#interface g0/0/1
Home-Gateway(config-if)#description LAN-Gateway
Home-Gateway(config-if)#ip address 200.23.16.1 255.255.255.0
Home-Gateway(config-if)#no shutdown
```

Step 3B: Re-configure LAN as Private Network

```
Home-Gateway(config)#ip dhcp excluded-address 200.23.16.1|
Home-Gateway(config)#ip dhcp pool LAN_POOL
Home-Gateway(dhcp-config)#network 200.23.16.0 255.255.255.0
Home-Gateway(dhcp-config)#default-router 200.23.16.1
Home-Gateway(dhcp-config)#dns-server 8.8.8.8
Home-Gateway(dhcp-config)#exit
```

Step 3C: Force PCs to renew IP

```
Home-Gateway(dhcp-config)#exit
Home-Gateway(config)#interface g0/0/0|
Home-Gateway(config-if)#description WAN-Link
Home-Gateway(config-if)#ip address 203.0.113.2 255.255.255.252
Home-Gateway(config-if)#no shutdown
Home-Gateway(config-if)#exit
Home-Gateway(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.1
Home-Gateway(config)#interface g0/0/1
Home-Gateway(config-if)#ip address 192.168.1.1 255.255.255.0
Home-Gateway(config-if)#exit
Home-Gateway(config)#no ip dhcp pool LAN_POOL
Home-Gateway(config)#ip dhcp pool PRIVATE_LAN
Home-Gateway(dhcp-config)#network 192.168.1.0 255.255.255.0
Home-Gateway(dhcp-config)#default-router 192.168.1.1
Home-Gateway(dhcp-config)#dns-server 8.8.8.8
Home-Gateway(dhcp-config)#exit
Home-Gateway(config)#%DHCPD-4-PING_CONFLICT: DHCP address conflict:  server pinged 192.168.1.1.
%DHCPD-4-PING_CONFLICT: DHCP address conflict:  server pinged 192.168.1.1.
```

Step 3D: Configure NAT Overload (PAT)

```
Home-Gateway(config)#interface g0/0/1
Home-Gateway(config-if)#ip nat inside
Home-Gateway(config-if)#exit
Home-Gateway(config)#interface g0/0/0
Home-Gateway(config-if)#ip nat outside
Home-Gateway(config-if)#exit
Home-Gateway(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Home-Gateway(config)#ip nat inside source list 1 interface g0/0/0 overload
Home-Gateway(config)#|
```

Verification: Open Command Prompt on PC-A. Ping 8.8.8.8.

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
|
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task 4

```
Home-Gateway>enable
Home-Gateway#show ip nat translations
Home-Gateway#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Home-Gateway(config)#ipv6 unicast-routing
Home-Gateway(config)#interface g0/0/1
Home-Gateway(config-if)#ipv6 address 2001:DB8:ACAD:1::1/64
Home-Gateway(config-if)#no shutdown
Home-Gateway(config-if)#
```

IPv6 Configuration

☒ Automatic

☐ Static

IPv6 request successful.

IPv6 Address

2001:DB8:ACAD:1:290:2BFF:FEA5:D25D

/ 64

Link Local Address

FE80::290:2BFF:FEA5:D25D

Default Gateway

FE80::201:42FF:FE01:3D02

DNS Server